Turkey, and New Zealand reported an overall survival rate of 56% (25 of 45 patients).[2] Eighty-three percent of HI neonates treated with systemic retinoids survived, whereas the long-term survival was only 24% for those who were not given oral retinoids.[2]

Here we report the outcomes for HI in the Japan population. For a clinical survey of HI in Japan, we distributed questionnaires to 904 dermatologic or pediatrics institutes or hospitals throughout Japan in 2010 and received responses from 564 institutes or hospitals (62.4%). Clinical data between 2005 and 2010 were obtained for 16 HI patients. In total, there were 13 survivors (81.3%) and 3 deceased (18.7%). The patient's sex was reported for 14 patients: 8 male and 6 female. Systemic retinoids were administered to 12 patients, 11 of whom survived (91.7%), whereas only 2 of the 4 patients (50%) who did not receive oral retinoids survived. Ten of 16 patients (62.5%) received intensive care in a neonatal intensive care unit (NICU).

Systemic retinoids and administration in the NICU were considered to contribute to relatively good outcomes for HI patients in the Japanese population.

The effects of oral retinoids in HI remains unproven,[3] and as suggested in the commentary by Milstone and Choate,[1] the early introduction of overall intensive therapy might have contributed to the better outcomes of the HI babies who were given oral retinoids.

In conclusion, we consider that, due to intensive neonatal care and, probably, to the early introduction of oral retinoids, HI outcomes have improved also in the Japanese population. In long-term HI survivors, epidermal keratinocytes might regain normal differentiation, and this restoration of differentiation is likely to be associated with the improvement of the skin symptoms in HI survivors.[4] From this fact and the results of our clinical survey, we agree with the commentary by Milstone and Choate[1] on the point that recognition of spontaneous improvement of the HI phenotype lowers the psychological hurdle and provides justification for intensive neonatal care that can improve the outcomes of HI patients.

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Research letters

Retronychia in children, adolescents, and young adults: A case series

To the Editor: The term *retronychia* describes the embedding of the proximal nail plate (NP) into the proximal nail fold (PNF) and has been reported mainly in adults.[1] We report here a case series of retronychia in children younger than 12 years of age, adolescents aged 13 to 18 years, and young adults aged 19 to 24 years.

All cases were discussed by dermatologists from 4 different centers. All patients gave informed consent for photographs and were followed

Fig 1. Retronychia: erythematous and edematous proximal nail fold, with a pyogenic granuloma protruding below the cuticle and a yellowish onycholytic nail plate.

Fig 2. Dystrophic nail following retronychia: disappearing nail bed, which impairs normal nail regrowth, is evident. The picture was taken 16 months after nail avulsion.

up long enough to document complete nail regrowth.

Fifteen patients were referred to the dermatologist because of pain and impaired walking, with a PNF that was erythematous and swollen, and stated that the NP had become yellow and had stopped growing (Fig 1). Granulation tissue below the cuticle was evident in 4 patients. Only 1 patient reported trauma as a precipitating event, 3 reported jogging or dancing as a causative factor, and 3 attributed the problem to uncomfortable shoes. Seven denied traumas but wore soft footwear during sports.

Thirteen of 15 patients were treated with complete NP avulsion and follow-up showed complete soft tissue healing in 12 of the 13 patients. One patient had a recurrence after 7 months. Five of the 13 patients had permanent nail dystrophy, presenting as a thickened yellowish nail that grew extremely slowly (Fig 2).

Two patients were treated with clobetasol propionate 0.05% ointment after refusing surgical intervention and suffering recurrence, 1 after 7 months and 1 after 12 months. Recurrences were retreated with a new cycle of clobetasol propionate 0.05% ointment in 2 patients and a new NP avulsion in 1 patient.

Our study shows that the clinical features of retronychia in young patients are the same as of those reported in adults and supports the theory that retronychia is more often caused by persistent minor trauma that interrupts the continuity between the nail matrix and the NP, as in onychomadesis, but in retronychia the new nail growing from the matrix pushes the old one upward, causing inflammation of the PNF and interrupting the longitudinal growth of the nail.[2,3]

We affirm that complete NP avulsion is the treatment of choice, leading to prompt healing, usually without recurrence, even though permanent nail dystrophy is frequent (5 of 15 patients in our series, 33%).

In patients who refuse surgery, clobetasol ointment may be helpful, even though recurrence may occur.

The English literature suggests that retronychia is a rare condition, but our experience is that it is less rare than publications suggest.[1-5] To the best of our knowledge, only 12 cases in patients younger than 24 years of age have been reported,[1-5] with the youngest being a 13-year-old girl.[3] It is unclear why retronychia is less frequent in children, adolescents, and young adults given that they wear soft shoes, play sports, and sustain more trauma than adults. We can hypothesize that this is due to a faster growth rate of their NP compared to that in adults.

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Pruritus severity in patients with psoriasis is not correlated with psoriasis disease severity

To the Editor: Pruritus has been reported in the majority of psoriatic patients, with 85% affected, and 77% having pruritus daily.[1,2] In a recent study, 80% of psoriatics reported that pruritus was the most important and severe symptom.[3] Pruritus has also been found to have a significant impact on the quality of life of psoriatics, with regards to falling asleep, and poorer psychosocial well-being, with patients found to exhibit feelings of stigmatization and depressive symptoms.[1,3,4] There is significant unmet need for effective treatment of the pruritus of psoriatics. In all, 45% of patients reported that no treatment relieved their pruritus, and only 8% of patients on topical corticosteroids found their pruritus was reduced.[1] Despite this, no therapies that specifically treat pruritus exist. Indeed, few clinical trials have assessed pruritus, and it is not part of any validated psoriasis scoring indices such as the Psoriasis Area Severity Index (PASI) or Investigator Global Assessment.

We report baseline data of 157 psoriatics enrolling for a clinical trial (EudraCT number: 2011-004640-21). These patients had mild to moderate psoriasis according to Investigator Global Assessment and a mean modified PASI (mPASI) score of 8.9. We found nearly all patients (97.5%) had pruritus; 68.8% had at least moderate pruritus (visual analog scale [VAS] score ≥ 40 mm) and 33.8% severe pruritus (VAS score ≥ 70 mm). This agrees with reports in the literature that suggest that 70% of psoriatics have at least moderate pruritus.[1]

Conflicting reports exist in the literature as to whether pruritus is associated with disease severity.[2,4,5] We thus explored whether there was a relationship between pruritus, measured by VAS, and psoriasis disease severity, measured by mPASI. mPASI primarily measures inflammatory-mediated aspects of psoriasis and does not directly measure pruritus. The lack of measurement of pruritus in the PASI scale should reduce bias when comparing the 2 indices. Fig 1 shows a scatterplot of pruritus severity (VAS) against psoriasis disease severity (mPASI) for the patients in the study. We found no evidence of any clinically meaningful correlation between pruritus severity and psoriasis disease severity. The coefficient of determination (R^2) value was 3.8%, meaning that only 3.8% of the variability in pruritus severity can be explained by its relationship to disease severity as measured by mPASI.

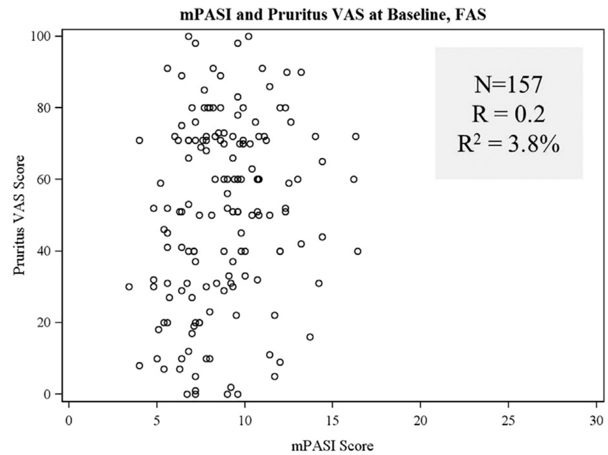

Fig 1. Lack of correlation between modified Psoriasis Area and Severity Index (*mPASI*) and pruritus visual analog scale (*VAS*) scores of study patients with psoriasis. Pruritus VAS scores, measuring pruritus severity, and mPASI scores, measuring psoriasis disease severity, were recorded in 157 patients at baseline enrolling for a clinical trial. The Psoriasis Area and Severity Index (PASI) score was modified as the scalp and face were not assessed. In all, 160 patients were enrolled in the trial, but 3 patients were excluded from the correlation analysis as they did not have properly recorded VAS or mPASI scores. The correlation between VAS and mPASI scores in these patients was measured through the coefficient of determination (R^2). A lack of correlation between VAS and mPASI scores was found. The R^2 value was 3.8%, meaning that only 3.8% of the variability in pruritus severity could be explained by its relationship to psoriasis disease severity. *FAS*, Full analysis set.

As mPASI measures inflammatory-mediated aspects of psoriasis, this lack of correlation suggests that treating pruritus in patients with psoriasis with anti-inflammatory agents may be a suboptimal approach. Current treatments for patients with mild-moderate psoriasis such as topical corticosteroids (glucocorticoids) and vitamin D analogs focus on treating the inflammatory component of psoriasis, with the intent that this will also treat any pruritus. Surprisingly, given the importance of pruritus to patients with psoriasis, the impact of these agents on